Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lab Cryptography Toolkit

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Crypto CTF Toolkit - Python Essentials

Method / Function	What it does
`.encode()`	Converts a string into bytes
`.decode()`	Converts bytes into a readable string
`bytes([...])`	Builds a bytes object from integers
`bytearray()`	Creates a mutable byte sequence
`int.from_bytes()`	Converts bytes to an integer
`int.to_bytes()`	Converts an integer to bytes
`bytes_to_long()`	Converts bytes to a large integer
`long_to_bytes()`	Converts a large integer to bytes

Crypto CTF Toolkit - Encoding

Method / Function	What it does
`bytes.fromhex()`	Converts hex strings into bytes
`binascii.unhexlify()`	Converts hex into bytes
`binascii.hexlify()`	Converts bytes into hex
`base64.b64decode()`	Decodes Base64-encoded data
`base64.b64encode()`	Encodes data in Base64

Crypto CTF Toolkit - Byte-wise Operations

Method / Function	What it does
`zip()`	Iterates over byte sequences in parallel
`^`	XOR operator - performs bitwise XOR on integers
`ord()`	Converts a character to an integer
`chr()`	Converts an integer to a character

Crypto CTF Toolkit – Rule of Thumb

> Crypto operates on bytes. Humans read strings. Math prefers integers.

Strings (Human-readable)

Examples:

- ASCII strings - "FLAG{CRYPTO_IS_FUN}"
- hex strings - 4d79537472696e67
- base64 strings - U0VMRUNUIFBBQ0s=

> Must be converted to bytes before crypto operations!

Crypto CTF Toolkit – Rule of Thumb

> Crypto operates on bytes. Humans read strings. Math prefers integers.

Bytes (Crypto-native)

Examples: b'\x46\x4c\x41\x47', b'Hello World'

Used for:

- Bitwise operations e.g. XOR, AND, NOT
- Encryption / decryption
- Hashing